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PH3102 Quantum Mechanics

End-semester Examination

Instructor: Dr. Siddhartha Lal

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Instructions: Answer all questions for full marks. Put your name and roll no. on both Question paper as well as the Answer script. and hand in both of them.

Date: 6th December, 2024

Full marks: 50

Time: 2.5 hours

1 Squeezed states. [7 marks]

For the 1D simple harmonic oscillator, define an operator $Q = \mu a + \nu a^\dagger$, where a and $a^\dagger$ are the lowering and raising operators respectively:

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{ip}{m\omega} \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{ip}{m\omega} \right), \quad [a, a^\dagger] = 1. \quad (1)$$

and where the complex numbers μ and ν satisfy the constraint $|\mu|^2 - |\nu|^2 = 1$.

(a) Show that $[Q, Q^\dagger] = 1$. [1 mark]

(b) Express the operators a and $a^\dagger$ in terms of Q and $Q^\dagger$. [1 mark]

(c) Now, consider an eigenstate $|\alpha\rangle$ of the operator Q , such that $Q|\alpha\rangle = \alpha|\alpha\rangle$ where $\alpha \in \mathbb{C}$. For the state $|\alpha\rangle$, and using the result of part (a), compute the expectation values $\langle x \rangle$ and $\langle x^2 \rangle$ for the position operator. [3 marks]

(d) Show that $\Delta x = \sqrt{\frac{\hbar}{2m\omega}} |\mu - \nu|$, where $|z| = \sqrt{zz^*}$ for a complex number z . [2 marks]

Note that such a state is called a "squeezed state" if Δx is less than its "natural" value $\sqrt{\frac{\hbar}{2m\omega}}$.

2 Density matrix and the Liouville-von Neumann equation. [12 marks]

Consider an arbitrary state $|\psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle$ for a spin-1/2 object, where the states $|\uparrow\rangle$ and $|\downarrow\rangle$ are the eigenstates of σ^z with eigenvalues 1/2 and -1/2 respectively. Such a state is placed in a magnetic field B , so that it evolves according to the Hamiltonian

$$H = B\sigma^z. \quad (2)$$

(a) Use the time evolution operator to find the time-evolved state $|\psi(t)\rangle$ at time t . [2 marks]

(b) Find the corresponding density matrix $\rho(t)$ at time t , where a density matrix is defined as $\rho = |\psi\rangle\langle\psi|$. [3 marks]

(c) Calculate the commutator $[H, \rho]$ between $\rho(t)$ and the Hamiltonian H . You can use the 2×2 matrix representation of ρ and H for this. [3 marks]

(d) By comparing the respective expressions, derive a relation between $d\rho/dt$ and $[H, \rho]$? This relation is called the Liouville-von Neumann equation. [Here, we consider the derivative of a matrix as simply element-wise differentiation.] [2 marks]

(e) What happens to the commutator when $b = 0$? What does this mean for the density matrix according to the Liouville-von Neumann equation? [2 marks]

3 Variational argument for Yukawa potential. [12 marks]

Consider the three dimensional attractive Yukawa potential:

$$V(r) = -A \frac{e^{-\lambda r}}{r}. \quad (3)$$

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where the dimensional constant $A > 0$ and the inverse distance scale $\lambda > 0$ sets the range of the potential (beyond which the potential falls off very quickly).

(a) Compute the expectation value of the Hamiltonian $H = -(\hbar^2/2m)\nabla^2 + V(r)$ for the normalised variational wavefunction

$$\psi_\alpha(r) = \sqrt{\frac{\alpha^3}{\pi}} e^{-\alpha r} \quad (4)$$

where the variational parameter α corresponds to a dimensionful constant a that sets a characteristic scale of the problem (e.g., for the case of $\lambda \rightarrow 0$, the Coulomb potential, $\alpha \propto a_0^{-1}$ the inverse Bohr radius and $A = e^2/4\pi\epsilon_0$). This will give you the variational energy $E(\lambda, \alpha)$. [6 marks]

Hint: Take the angular momentum quantum number $l = 0$ in the relation for the operator

$$\nabla^2 \equiv \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \quad (5)$$

(b) Minimise the variational energy $E(\alpha, \lambda)$ with respect to the variational parameter α to obtain the best variational estimate of the ground state energy E_{min} . State clearly the relation you obtain between α and λ as an outcome of the variational calculation. [4 marks]

(c) Show that there exists a relationship for the parameter λ such that E_{min} can accommodate bound states, $E_{min} < 0$. What does it indicate for the repulsive case of $A < 0$? [2 marks]

4 Anisotropic spin system. [12 marks]

(a) Two spins S_1 and S_2 are interacting with one other through the antiferromagnetic Ising spin exchange Hamiltonian

$$H_1 = J_{\parallel} S_1^z S_2^z \quad J_{\parallel} > 0 \quad (6)$$

Compute the eigenvectors and eigenvalues of the Hamiltonian H_1 . [2 marks]

(b) Now consider two spins S_1 and S_2 that are interacting with one other through the antiferromagnetic anisotropic spin exchange Hamiltonian

$$H_2 = J_{\perp} (S_1^x S_2^x + S_1^y S_2^y) + J_{\parallel} S_1^z S_2^z \quad J_{\parallel} > 0, J_{\perp} > 0 \quad (7)$$

Again, compute the eigenvectors and eigenvalues of the Hamiltonian H_2 . How has the presence of the extra term in H_2 , i.e., the term proportional $J_{\perp}$, affected the energy eigenvalues of H_2 with respect to those obtained for H_1 ? [7 marks]

(c) What happens to the eigenspectrum of H_2 for the special case of $J_{\perp} = J_{\parallel}$? [1 mark]

(d) Without conducting any explicit calculations, sketch pictorially the energy spectrum for the ferromagnetic exchange Hamiltonian with $J_{\parallel}, J_{\perp} < 0$ and (i) $J_{\parallel} \neq J_{\perp}$, and (ii) $J_{\parallel} = J_{\perp}$. [2 marks]

5 Perturbing the 2D Harmonic Oscillator. [7 marks]

Consider the two dimensional harmonic oscillator $H_0 = \frac{1}{2m}(p_x^2 + p_y^2) + \frac{m\omega^2}{2}(x^2 + y^2)$, to which we apply the perturbation $V = \lambda xy$.

(a) Write down the energy spectrum of H_0 . For this, use the good quantum numbers that label the eigenstates of H_0 . [2 marks]

(b) Compute the lowest order correction to the ground state of H_0 due to V . [2 marks]

(c) Compute the lowest order correction to the first excited state(s) of H_0 due to V . [3 marks]

Hint: Consider carefully any degeneracies in these states.